

State-space Gromov–Hausdorff convergence of truncated tensor-product Camporesi–Higuchi spectral triples on $S^3 \times S^3$

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May 2026 (preprint)

Abstract

We prove that the Connes–van Suijlekom truncations of the tensor-product Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ on $S^3 \times S^3$ at two (possibly distinct) focal lengths $\lambda_a, \lambda_b > 0$ converge to the continuum tensor-product triple in van Suijlekom’s state-space Gromov–Hausdorff distance, with explicit qualitative-rate bound $\Lambda(\mathcal{T}_{n_{\max},a} \otimes \mathcal{T}_{n_{\max},b}, \mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}) \leq C_3^{(2)} \cdot (1 + 2\sqrt{2}) \cdot \max(\gamma_{n_{\max},a}/\lambda_a, \gamma_{n_{\max},b}/\lambda_b)$, where $C_3^{(2)} \leq \sqrt{2}$ is a closed-form Lipschitz comparison constant from the (tight) triangle bound on the Connes–Marcolli joint Leibniz rule, and $\gamma_n = (4/\pi) \log n/n + O(1/n)$ inherits from the single-factor result of [20]. The proof proceeds factor-by-factor through tensor extensions of the five single-factor lemmas (operator system, central spectral Fejér kernel, Lipschitz comparison, Berezin reconstruction, propinquity assembly). The two technical risks specific to the tensor product — a closed-form Lipschitz comparison constant on the full operator system, and a rigorous bound on the cross-Stein–Weiss height term — both close via the triangle bound on the joint Leibniz rule. (An earlier Pythagorean refinement claiming $C_3^{(2)} < 1$ has been *withdrawn*: the graded anticommutation does *not* make the two Leibniz terms operator-norm orthogonal — on the actual Camporesi–Higuchi harmonics they add constructively, so the triangle bound is tight and $C_3^{(2)} \nearrow \sqrt{2}$, exceeding 1 for cutoffs beyond (3,3). The convergence theorem is unaffected, since $\sqrt{2}$ is a finite constant.) The strictly stronger Latrémolière propinquity is *not* claimed: the dual-reach (reach_P) step of the joint assembly inherits the single-factor reach_P named gap of Papers 38/40 (the invalid cb-norm dual-reach), so the object established is van Suijlekom state-space Gromov–Hausdorff convergence. The result extends [6, 15, 20] to the simplest non-abelian tensor case, and closes a tensor-product state-space-GH case that the published literature on products of metric spectral triples [1, 7, 8, 14] does not cover; existing tensor-product propinquity results require the second factor to be finite-dimensional, AF, abelian, or to reduce to a single-character / Hopf- S^1 structure.

MSC2020 classifications: 58B34 (noncommutative geometry), 46L87 (noncommutative Riemannian and metric geometry), 46L52 (noncommutative function spaces), 22E45 (representations of compact Lie groups), 46L06 (tensor products of operator systems).

Keywords: spectral triple, tensor product, Connes–Marcolli graded Dirac, quantum Gromov–Hausdorff propinquity, Camporesi–Higuchi Dirac, operator system, Berezin reconstruction, graded Leibniz rule, $SU(2) \times SU(2)$.

*Independent researcher. jloutey@gmail.com. This work was developed in an AI-augmented agentic workflow with Anthropic’s Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author’s responsibility.

1 Introduction

The Connes–van Suijlekom [6] framework of truncated operator systems and the subsequent propinquity-convergence results of Hekkelman [9], Hekkelman–McDonald [10], Leimbach–van Suijlekom [15], and the present author [20] establish quantitative Gromov–Hausdorff convergence for spectral truncations of a *single* compact Riemannian spectral triple, with the abelian and $SU(2)$ cases now closed at qualitative rate. A natural and operationally important next question is whether the convergence behaviour transfers to *tensor products* of two such truncated triples.

The published propinquity tensor-product literature does not cover this case in its central form. Latrémolière [14] closes the *almost-commutative* case in which one factor is an infinite-dimensional Riemannian spectral triple and the other is finite-dimensional. Farsi–Latrémolière [7] prove spectral continuity for collapse of products of spectral triples in which at least one factor is abelian. Farsi–Latrémolière [8] treat continuity of a *single* metric spectral triple along an analytic path of Riemannian metrics — a setting distinct from the discrete-truncation product considered here. Aguilar [1] addresses tensor products $C(X) \otimes A$ where one factor is commutative and the other AF. Hekkelman–McDonald [10] treats a noncommutative integral on a single spectrally truncated triple, not a product. The published thread covers Riemannian-by-finite, Riemannian-by-AF, Riemannian-by-abelian, $C(X)$ -by-AF, single-triple-along-paths, and single-truncation. None of these covers two infinite non-abelian Riemannian metric spectral triples.

The present paper closes that gap in the simplest physically natural non-Kähler case: two copies of the round $S^3 = SU(2)$ Camporesi–Higuchi triple [4], each in its own focal-length scaling, tensored together via the Connes–Marcolli graded convention.

Let $\mathcal{T}_{S^3}^\lambda = (C^\infty(S^3), L^2(S^3, \Sigma), \lambda^{-1}D_{\text{CH}})$ denote the round Camporesi–Higuchi spectral triple at *focal length* $\lambda > 0$, i.e. the unit-round triple with the Dirac operator rescaled so that the spectral gap is λ^{-1} rather than 1. (In the GeoVac hydrogenic context this corresponds to placing two electrons in distinct hydrogenic shells with focal lengths $\lambda_a = n_a/Z$ and $\lambda_b = n_b/Z$.) Form the tensor product

$$\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b} := (C^\infty(S^3) \otimes C^\infty(S^3), L^2(S^3, \Sigma) \otimes L^2(S^3, \Sigma), D_{a,b}), \quad (1)$$

with Connes–Marcolli graded composed Dirac

$$D_{a,b} = \lambda_a^{-1}D_{\text{CH}}^{(a)} \otimes I_b + \gamma_a \otimes \lambda_b^{-1}D_{\text{CH}}^{(b)}, \quad (2)$$

where γ_a is the chirality grading on $L^2(S^3, \Sigma)$. The combined object has KO-dimension $3+3=6 \pmod{8}$. Truncate each factor independently at cutoffs $n_{\max,a}, n_{\max,b}$ and form the tensor-product truncated triple $\mathcal{T}_{n_{\max,a}} \otimes \mathcal{T}_{n_{\max,b}}$.

Theorem 1.1 (Main theorem; cf. §3). *The truncated tensor-product triples $\mathcal{T}_{n_{\max,a}} \otimes \mathcal{T}_{n_{\max,b}}$ converge to $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ in van Suijlekom’s state-space Gromov–Hausdorff distance, with bound*

$$\Lambda(\mathcal{T}_{n_{\max,a}} \otimes \mathcal{T}_{n_{\max,b}}, \mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}) \leq C_3^{(2)}(n_{\max,a}, n_{\max,b}) \cdot (1 + 2\sqrt{2}) \cdot \max\left(\frac{\gamma_{n_{\max,a}}}{\lambda_a}, \frac{\gamma_{n_{\max,b}}}{\lambda_b}\right), \quad (3)$$

where γ_n is the central spectral Fejér moment of [20] satisfying $\gamma_n = (4/\pi) \log n/n + O(1/n)$, and the joint Lipschitz comparison constant is

$$C_3^{(2)}(n_{\max,a}, n_{\max,b}) \leq \sup_{\substack{2 \leq N_a \leq n_{\max,a}+1 \\ 2 \leq N_b \leq n_{\max,b}+1}} \sqrt{\frac{((N_a - 1) + (N_b - 1))^2}{N_a^2 + N_b^2 - 2}}, \quad (4)$$

bounded by $\sqrt{2}$ at every finite $(n_{\max,a}, n_{\max,b})$ and approaching $\sqrt{2}$ from below as both cutoffs grow (this is the tight triangle bound; the convergence is driven by $\gamma_n \rightarrow 0$, not by the constant).

The proof is a factor-by-factor lift of the five single-factor lemmas of [20] to the tensor product, supplemented by two new ingredients specific to the product:

- a closed-form Lipschitz comparison estimate from the (tight) operator-norm triangle bound $\|A + B\|_{\text{op}} \leq \|A\| + \|B\|$ on the two Connes–Marcolli Leibniz terms (Lemma 4.6); and
- an explicit Stein–Weiss-type bound on the joint Lipschitz-distortion height of the tunnel map, again via the triangle bound applied to the joint commutator (Lemma 4.11).

(An earlier draft used a *Pythagorean* refinement of these two bounds, claiming $C_3^{(2)} < 1$; that refinement is withdrawn — the graded anticommutation does not produce operator-norm orthogonality — and replaced throughout by the rigorous triangle bound, giving $C_3^{(2)} \leq \sqrt{2}$; see Remark 4.5.)

The mechanism is unified: the same KO-dim-6 graded structure that defines the Connes–Marcolli composed Dirac splits both the joint Lipschitz constant (R1) and the joint height-distortion bound (R2) into Leibniz terms controlled by the triangle bound. Each of the five single-factor lemmas of [20] extends factor-by-factor with no novel analytical machinery.

Why this is the *easy* case. The same-manifold tensor product treated here is the most accessible of three plausible tensor-product extensions of [20]. The cross-manifold case $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Bargmann}}$, in which the second factor is a holomorphic Hardy-sector spectral triple of the type used in the Bargmann–Segal lattice construction [19], is structurally obstructed at the framework level: the S^3 factor is a second-order Riemannian spinor Dirac, while the S^5 Bargmann factor is a first-order complex-analytic Euler operator on a Hardy sector and does not fit standard NCG Dirac axioms. There is no published prescription for tensoring a Riemannian spinor Dirac with a Hardy-sector first-order complex operator in the propinquity framework, and the obstruction is the four-layer Coulomb/HO asymmetry of [19] §V resurfacing at the spectral-triple level. A separate variant — $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Riem}}$ with both factors Riemannian — is mathematically clean (KO-arithmetic $3 + 5 = 8 \equiv 0 \pmod{8}$) but requires propinquity convergence for the rank-2 $\mathcal{T}_{S^5}^{\text{Riem}}$ as a separate prerequisite, which is not available in the literature. We address only the same-manifold case in the present paper, and discuss the obstructions to the cross-manifold extensions in §6.

Outline. Section 2 sets up the rescaled Camporesi–Higuchi spectral triple at focal length λ , the truncations, and the tensor-product Connes–Marcolli graded structure. Section 3 states the main theorem. Section 4 proves the five tensor lemmas. Section 5 reports the numerical verification panel. Section 6 places the result in the Connes–van Suijlekom / Latrémolière / Farsi–Latrémolière / Hekkelman–McDonald lineage and discusses obstructions to the cross-manifold extensions. Section 7 lists open questions. Section 8 concludes.

2 The setting

Convergence convention and descope note. The object established here is van Suijlekom’s *state-space Gromov–Hausdorff distance* (the convention of Paper 38 [20]), *not* the strictly stronger Latrémolière propinquity [12] (Adv. Math. **404** (2022), Paper No. 108393, preprint [arXiv:1811.10843](#)). The bound symbol $\Lambda(\cdot, \cdot)$ below denotes the state-space GH distance. The joint assembly (Lemma 4.11) bounds the dual reach $\text{reach}_{P_{n_{\max},a} \otimes P_{n_{\max},b}}$ by inheriting the single-factor reach_P estimate; that single-factor step is the invalid cb-norm dual-reach that descope Papers 38/40 from Latrémolière propinquity to state-space GH, so the joint result inherits the same named gap and is stated at the state-space-GH level. The rate and the joint Lipschitz constant $C_3^{(2)} \leq \sqrt{2}$ (Lemma 4.6, triangle bound) are unaffected by this distinction.

2.1 Single-factor recall: the rescaled S^3 Camporesi–Higuchi triple

For $\lambda > 0$, the λ -scaled Camporesi–Higuchi spectral triple on the unit-round $S^3 = \text{SU}(2)$ is

$$\mathcal{T}_{S^3}^\lambda = (C^\infty(S^3), L^2(S^3, \Sigma), \lambda^{-1} D_{\text{CH}}), \quad (5)$$

with D_{CH} as in [20] §2.1, eigenvalues $\sigma(\lambda^{-1} D_{\text{CH}}) = \{\pm \lambda^{-1}(n + 1/2) : n \in \mathbb{N}\}$ on each chirality sector. The truncated triple at cutoff $n_{\max}$ is $\mathcal{T}_{n_{\max}}^\lambda = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, \lambda^{-1} D_{n_{\max}})$, with the truncated operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ and the truncated Dirac $D_{n_{\max}} = P_{n_{\max}} D_{\text{CH}} P_{n_{\max}}$. We will write $\mathcal{T}^\lambda$ when the dependence on λ matters; the rescaling multiplies single-factor Connes distances and propinquity bounds by λ , but is otherwise transparent for our purposes.

2.2 The tensor-product spectral triple

Let $\lambda_a, \lambda_b > 0$. Define

$$\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b} := (\mathcal{A}_{a,b}, \mathcal{H}_{a,b}, D_{a,b}, \gamma_{a,b}, J_{a,b}), \quad (6)$$

with

$$\mathcal{A}_{a,b} = C^\infty(S^3) \otimes C^\infty(S^3) = C^\infty(S^3 \times S^3), \quad (7)$$

$$\mathcal{H}_{a,b} = L^2(S^3, \Sigma) \otimes L^2(S^3, \Sigma), \quad (8)$$

$$D_{a,b} = \lambda_a^{-1} D_{\text{CH}}^{(a)} \otimes I_b + \gamma_a \otimes \lambda_b^{-1} D_{\text{CH}}^{(b)}, \quad (9)$$

$$\gamma_{a,b} = \gamma_a \otimes \gamma_b, \quad (10)$$

$$J_{a,b} = J_a \otimes J_b \quad (11)$$

in the standard Connes–Marcolli graded tensor product convention [5, 16]. The crucial structural identity is

$$\{D_{\text{CH}}^{(a)} \otimes I_b, \gamma_a \otimes D_{\text{CH}}^{(b)}\} = (D_{\text{CH}}^{(a)} \gamma_a + \gamma_a D_{\text{CH}}^{(a)}) \otimes D_{\text{CH}}^{(b)} = 0, \quad (12)$$

since $\{D_{\text{CH}}^{(a)}, \gamma_a\} = 0$ on the unit-round S^3 Dirac. This anticommutation of the two *Dirac* summands is what gives the clean two-term graded Leibniz rule (18) (it does *not*, however, make the two resulting *commutator* terms operator-norm orthogonal — see Remark 4.5); the joint commutator is then controlled by the triangle bound, which suffices to make the tensor-product propinquity convergent.

The truncated tensor-product triple at cutoffs $(n_{\max,a}, n_{\max,b})$ is

$$\mathcal{T}_{n_{\max,a}}^{\lambda_a} \otimes \mathcal{T}_{n_{\max,b}}^{\lambda_b} := (\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}, \mathcal{H}_{n_{\max,a}} \otimes \mathcal{H}_{n_{\max,b}}, P_{n_{\max,a}} \otimes P_{n_{\max,b}} \cdot D_{a,b} \cdot P_{n_{\max,a}} \otimes P_{n_{\max,b}}), \quad (13)$$

with the algebraic tensor-product operator system $\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}} \subset M_{\dim \mathcal{H}_{n_{\max,a}}}(\mathbb{C}) \otimes M_{\dim \mathcal{H}_{n_{\max,b}}}(\mathbb{C})$ defined as the linear span of Kronecker products $M_a \otimes M_b$ with $M_\bullet \in \mathcal{O}_{n_{\max}, \bullet}$.

3 Main theorem

Theorem 3.1 (Main theorem). *The truncated tensor-product Camporesi–Higuchi spectral triples converge to their continuum tensor-product limit in van Suijlekom’s state-space Gromov–Hausdorff distance (the strictly stronger Latrémolière propinquity is not claimed; see the convergence convention below):*

$$\Lambda(\mathcal{T}_{n_{\max,a}}^{\lambda_a} \otimes \mathcal{T}_{n_{\max,b}}^{\lambda_b}, \mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}) \leq C_3^{(2)}(n_{\max,a}, n_{\max,b}) \cdot (1 + 2\sqrt{2}) \cdot \max\left(\frac{\gamma_{n_{\max,a}}}{\lambda_a}, \frac{\gamma_{n_{\max,b}}}{\lambda_b}\right), \quad (14)$$

with

$$C_3^{(2)}(n_{\max,a}, n_{\max,b}) := \sup_{\substack{2 \leq N_a \leq n_{\max,a}+1 \\ 2 \leq N_b \leq n_{\max,b}+1}} \sqrt{\frac{((N_a - 1) + (N_b - 1))^2}{N_a^2 + N_b^2 - 2}} \leq \sqrt{2} \quad (15)$$

at every finite $(n_{\max,a}, n_{\max,b})$, and $C_3^{(2)} \nearrow \sqrt{2}$ as both cutoffs grow. (This is the tight triangle bound on the joint Leibniz rule; it is achieved with equality on a pair of top-shell raising harmonics, where the two Leibniz terms add constructively. The earlier closed form $\sqrt{((N_a - 1)^2 + (N_b - 1)^2)/(N_a^2 + N_b^2)}$ rested on a Pythagorean operator-norm identity that does not hold for these operators; see Lemma 4.6.) The right-hand side of (14) tends to zero as $n_{\max,a}, n_{\max,b} \rightarrow \infty$, with rate $\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}) = O(\log n_{\max}/n_{\max})$ inherited from the single-factor central Fejér moment [20]. The asymptotic constant $4/\pi$ on each factor is preserved in the joint rate.

Proof. The proof assembles five tensor lemmas L1'-T through L5-T proved in §4, each of which lifts the corresponding single-factor lemma of [20] §3. Lemma 4.1 provides the operator-system substrate; Lemma 4.3 the joint central spectral Fejér kernel with closed-form Plancherel symbol; Lemma 4.6 the joint Lipschitz comparison constant via the (tight) triangle bound on the Connes–Marcolli graded Leibniz rule; Lemma 4.10 the joint Berezin reconstruction with the four properties needed for a Latrémolière tunneling pair; and Lemma 4.11 the state-space-GH assembly with explicit cross-Stein–Weiss height bound (modulo the inherited reach_P named gap; see the convergence convention). The factor $1 + 2\sqrt{2}$ in (14) is the sum of the joint reach contribution (bounded by $\max(\gamma)$, contributing 1) and the joint height contribution (bounded by $2\sqrt{2}\max(\gamma)$ on the unit-norm panel, contributing $2\sqrt{2}$). The λ^{-1} -rescalings of the single-factor Diracs propagate linearly through the bound and produce the $\max(\gamma/\lambda)$ -form of (14). $\square$

4 The five tensor lemmas

We work at unit focal length $\lambda_a = \lambda_b = 1$ throughout this section; the λ -rescaling is reinstated at the end of §4.5.

4.1 Lemma L1'-T: tensor-product operator system

Lemma 4.1 (L1'-T: tensor-product operator system substrate). *Let $\mathcal{O}_{n_{\max,a}}$ and $\mathcal{O}_{n_{\max,b}}$ be the truncated operator systems of [20, §3.1] & [20, Lemma 3.1]. The algebraic tensor product*

$$\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}} \subset M_{\dim \mathcal{H}_{n_{\max,a}}}(\mathbb{C}) \otimes M_{\dim \mathcal{H}_{n_{\max,b}}}(\mathbb{C}),$$

defined as the linear span of Kronecker products $M_a \otimes M_b$ with $M_a \in \mathcal{O}_{n_{\max,a}}$, $M_b \in \mathcal{O}_{n_{\max,b}}$, satisfies:

- (a) (*Dimension factorization.*) $\dim(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) = \dim(\mathcal{O}_{n_{\max,a}}) \cdot \dim(\mathcal{O}_{n_{\max,b}})$.
- (b) (*Propagation bound.*) $\text{prop}(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) \leq 2$, matching the single-factor Connes–van Suijlekom $\text{prop} = 2$ value [6, §4.2].
- (c) (*Cross-shell finite Connes distance.*) Under the chirality-mixing offdiagonal extension $D_{a,b}^{\text{offdiag}}$ defined factor-by-factor in the sense of [20, Lemma 3.1], every cross-shell pair of pure node-evaluation states has finite truncated joint Connes distance.

Proof. (a) The single-factor multiplier matrices $\{M_{NLM}\}$ are linearly independent in $M_{\dim \mathcal{H}_{n_{\max},\bullet}}(\mathbb{C})$ by construction ([20, §3.1]). The Kronecker product preserves linear independence: $\text{vec}(M_a \otimes M_b) = \text{vec}(M_a) \otimes \text{vec}(M_b)$, and tensors of linearly independent vectors are linearly independent. Hence $\dim(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) = \dim(\mathcal{O}_{n_{\max,a}}) \cdot \dim(\mathcal{O}_{n_{\max,b}})$.

(b) For any operator system $\mathcal{O}$ with $\text{prop}(\mathcal{O}) = p$, $\mathcal{O}^p = M_{\dim \mathcal{H}(\mathbb{C})}$. For $p_a = p_b = 2$, we have $(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}})^2 \supset \mathcal{O}_{n_{\max,a}}^2 \otimes \mathcal{O}_{n_{\max,b}}^2 = M_{\dim \mathcal{H}_{n_{\max,a}}(\mathbb{C})} \otimes M_{\dim \mathcal{H}_{n_{\max,b}}(\mathbb{C})} = M_{\dim \mathcal{H}_{a,b}(\mathbb{C})}$, so $\text{prop}(\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}) \leq 2$.

(c) The chirality-mixing offdiagonal extension $D_{a,b}^{\text{offdiag}}$ inherits the cross-shell-finite property factor-by-factor from [20, Lemma 3.1]. For a state pair $(\varphi_a \otimes \varphi_b, \psi_a \otimes \psi_b)$ in distinct shells, at least one factor differs in shell, and the single-factor finite-distance property on that factor produces a non-trivial structural multiplier bounding the supremum in the SDP definition of the joint truncated Connes distance. $\square$

Remark 4.2. As in [20, Remark 3.2], the offdiagonal extension is a metric-only device for finite-distance verification; the truthful $D_{a,b} = D_{\text{CH}}^{(a)} \otimes I + \gamma_a \otimes D_{\text{CH}}^{(b)}$ is what satisfies the KO-dim-6 chirality and real-structure axioms and is what underlies Lemmas 4.6–4.10.

4.2 Lemma L2-T: joint central spectral Fejér kernel

Let $K_{n_{\max,a}}^{(a)}$ and $K_{n_{\max,b}}^{(b)}$ denote the single-factor central spectral Fejér kernels on $\text{SU}(2)$ defined in [20, Definition 3.3]. Define the joint product kernel

$$K_{n_{\max,a}, n_{\max,b}}^{(a,b)}(g_1, g_2) := K_{n_{\max,a}}^{(a)}(g_1) K_{n_{\max,b}}^{(b)}(g_2) \quad \text{on } \text{SU}(2) \times \text{SU}(2). \quad (16)$$

Lemma 4.3 (L2-T: joint kernel construction and rate). *The joint central kernel $K^{(a,b)}$ has the following properties:*

- (a) *(Positivity and unit mass.)* $K^{(a,b)}(g_1, g_2) \geq 0$ on $\text{SU}(2) \times \text{SU}(2)$, and $\int (K^{(a,b)}) dg_1 dg_2 = 1$ in the product Haar measure.
- (b) *(Plancherel factorization.)* The joint Plancherel symbol factorizes: $\widehat{K}^{(a,b)}(j_1, j_2) = \widehat{K}^{(a)}(j_1) \cdot \widehat{K}^{(b)}(j_2)$, with each factor the closed-form $\widehat{K}^{(\bullet)}(j) = (2j+1)/Z_{n_{\max,\bullet}}$ for $j \leq j_{\max,\bullet}$ and zero otherwise.
- (c) *(Cb-norm factorization.)* $\|T_{K^{(a,b)}}\|_{\text{cb}}^{\text{cent}} = \|T_{K^{(a)}}\|_{\text{cb}}^{\text{cent}} \cdot \|T_{K^{(b)}}\|_{\text{cb}}^{\text{cent}} = 4/[(n_{\max,a}+1)(n_{\max,b}+1)]$.
- (d) *(Subadditive joint mass concentration.)* Define $\gamma_{n_{\max,a}, n_{\max,b}}^{(a,b)} := \int K^{(a,b)}(g_1, g_2) d_{S^3 \times S^3}((e, e), (g_1, g_2)) dg_1 dg_2$. Then

$$\gamma_{n_{\max,a}, n_{\max,b}}^{(a,b)} \leq \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}} \leq 2 \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}). \quad (17)$$

Proof. (a) The product of two non-negative unit-mass kernels is a non-negative unit-mass kernel under the product Haar measure. (b) The Plancherel decomposition of $L^2(G_1 \times G_2)$ is the product of the two single-factor Plancherel decompositions, and convolution by a central function on a product of compact groups acts diagonally on each isotypic component as the product of single-factor Fourier coefficients. (c) The Bożejko–Fendler equality $\|T_K\|_{\text{cb}} = \|\widehat{K}\|_{\ell^\infty}$ for central convolutions on amenable compact groups [3, 17] extends to product kernels via the standard tensor-product cb-norm identity $\|A \otimes B\|_{\text{cb}} = \|A\|_{\text{cb}} \|B\|_{\text{cb}}$ on the minimal tensor product [17, Ch. 8]. Substituting the single-factor result [20, Lemma 3.4(c)] gives the displayed value. (d) The product Riemannian metric on $S^3 \times S^3$ satisfies the triangle inequality $d_{S^3 \times S^3}((e, e), (g_1, g_2)) \leq d_{S^3}(e, g_1) + d_{S^3}(e, g_2)$ at every point. Integrating against $K^{(a,b)}(g_1, g_2) dg_1 dg_2$ and applying Fubini gives (17). $\square$

Remark 4.4. Property (c) shows that the joint cb-norm is one order faster than the single-factor rate: $\|T_{K^{(a,b)}}\|_{\text{cb}} = O(1/(n_{\max,a} n_{\max,b}))$ vs. single-factor $O(1/n_{\max})$. This is the structural reason joint convergence is well-behaved.

4.3 Lemma L3-T: joint Lipschitz comparison via the triangle bound on the graded Leibniz rule

This is the central new technical content of the paper. The Connes–Marcolli graded Dirac (9) on the KO-dim-6 graded module splits the joint commutator into two Leibniz terms (18); the operator-norm triangle bound on that sum is what controls the joint Lipschitz comparison constant.

Remark 4.5 (Withdrawn Pythagorean refinement). An earlier version of this lemma claimed the *Pythagorean* identity $\|A + B\|_{\text{op}}^2 = \|A\|^2 + \|B\|^2$ for the two Leibniz terms (citing the graded anticommutation $\{A, B\} = 0$), which would have given $C_3^{(2)} < 1$. That refinement is **false**: $\{A, B\} = 0$ does *not* imply operator-norm orthogonality (the elementary counterexample $A = \text{diag}(1, 0)$, $B = \text{diag}(0, 1)$ has $\{A, B\} = 0$ but $\|A + B\|^2 = 1 \neq 2 = \|A\|^2 + \|B\|^2$), and the specific factorisation that was invoked is invalid because the two Leibniz terms carry *different* second tensor factors ($M_g[D_{\text{CH}}^{(b)}, M_g]$ vs. $[D_{\text{CH}}^{(b)}, M_g]M_g$, which do not commute). A direct numerical test on the actual Camporesi–Higuchi harmonics (the supporting test below) finds the Pythagorean bound *maximally violated* ($\|A + B\|^2/(\|A\|^2 + \|B\|^2) = 2$ exactly at a pair of top-shell raising harmonics, where the two terms add constructively) and the *triangle* bound tight ($\|A + B\| = \|A\| + \|B\|$ there). The constant is therefore $C_3^{(2)} \nearrow \sqrt{2}$, not < 1 . This does not affect convergence: $\sqrt{2}$ is a finite constant, so the right-hand side of (14) still $\rightarrow 0$ as the cutoffs grow.

Lemma 4.6 (L3-T: joint Lipschitz comparison constant). *For every simple-tensor multiplier $M_{f \otimes g} = M_f \otimes M_g$ with $f, g \in C^\infty(S^3)$, the joint Connes–Marcolli commutator satisfies the Leibniz identity*

$$[D_{a,b}, M_f \otimes M_g] = [D_{\text{CH}}^{(a)}, M_f] \otimes M_g + (\gamma_a M_f) \otimes [D_{\text{CH}}^{(b)}, M_g], \quad (18)$$

and, by the operator-norm triangle inequality applied to the two Leibniz terms (which is tight — see Remark 4.5),

$$\|[D_{a,b}, M_f \otimes M_g]\|_{\text{op}} \leq \|[D_{\text{CH}}^{(a)}, M_f]\|_{\text{op}} \cdot \|M_g\|_{\text{op}} + \|M_f\|_{\text{op}} \cdot \|[D_{\text{CH}}^{(b)}, M_g]\|_{\text{op}}. \quad (19)$$

Combining (19) with the single-factor L3 estimate [20, Lemma 3.5] for each factor at level (N_a, N_b) in the Avery 3-Y harmonic basis gives the per-irrep ratio

$$\frac{\|[D_{a,b}, M_{N_a L_a M_a} \otimes M_{N_b L_b M_b}]\|_{\text{op}}^2}{\|Y_{N_a L_a M_a}^{(3)} \otimes Y_{N_b L_b M_b}^{(3)}\|_{\text{Lip}}^2} \leq \frac{((N_a - 1) + (N_b - 1))^2}{N_a^2 + N_b^2 - 2}. \quad (20)$$

Taking square roots and supping over the irrep grid gives the joint Lipschitz comparison constant $C_3^{(2)}(n_{\text{max},a}, n_{\text{max},b})$ of (15), which is bounded by $\sqrt{2}$ at every finite $(n_{\text{max},a}, n_{\text{max},b})$ and increases to $\sqrt{2}$ as the cutoffs grow.

Proof. The Leibniz identity (18) is a direct computation on Kronecker products:

$$[D_{a,b}, M_f \otimes M_g] = [D_{\text{CH}}^{(a)} \otimes I_b, M_f \otimes M_g] + [\gamma_a \otimes D_{\text{CH}}^{(b)}, M_f \otimes M_g],$$

with $[D_{\text{CH}}^{(a)} \otimes I_b, M_f \otimes M_g] = [D_{\text{CH}}^{(a)}, M_f] \otimes M_g$ (since I_b commutes with M_g), and $[\gamma_a \otimes D_{\text{CH}}^{(b)}, M_f \otimes M_g] = \gamma_a M_f \otimes [D_{\text{CH}}^{(b)}, M_g]$ (since $\gamma_a M_f \neq M_f \gamma_a$ in general but the chirality grading commutes with the scalar multiplier as a chirality-blind operator; in components, the chirality flip is absorbed into the γ_a factor on the left).

The bound (19) is the operator-norm triangle inequality $\|A + B\|_{\text{op}} \leq \|A\| + \|B\|$ applied to the two Leibniz terms $A = [D_{\text{CH}}^{(a)}, M_f] \otimes M_g$ and $B = (\gamma_a M_f) \otimes [D_{\text{CH}}^{(b)}, M_g]$, together

with the multiplicativity of the operator norm on simple tensors ($\|X \otimes Y\| = \|X\| \|Y\|$, and $\|\gamma_a M_f\| = \|M_f\|$ since γ_a is unitary). This bound is *tight*: on a pair of top-shell raising harmonics the two terms align so that $\|A + B\| = \|A\| + \|B\|$ exactly. There $\|A\| = \|B\|$, so $\|A + B\|^2 = (2\|A\|)^2 = 4\|A\|^2$ while the Pythagorean value is $\|A\|^2 + \|B\|^2 = 2\|A\|^2$ — a factor 2 violation of the would-be identity (see Remark 4.5 for why the graded anticommutation does *not* give operator-norm orthogonality). The numerical verification on the actual Camporesi–Higuchi harmonics — the joint commutator built from the real D_{CH} matrices and the graded multipliers at $(n_{\max,a}, n_{\max,b}) = (2, 2), (2, 3)$, with the operator norm computed by singular-value decomposition — finds, over every harmonic pair, the triangle ratio $\|A + B\|/(\|A\| + \|B\|)$ attaining its maximum 1 and the Pythagorean ratio $\|A + B\|^2/(\|A\|^2 + \|B\|^2)$ attaining its maximum 2, confirming both the tightness of (19) and the failure of the Pythagorean form (backing test `tests/test_paper39_triangle_tight.py`).

For the per-irrep ratio (20), the single-factor L3 estimate [20, Eq. (3.18)] gives $\|[D_{\text{CH}}^{(\bullet)}, M_{N_\bullet L_\bullet M_\bullet}]\|_{\text{op}} \leq (N_\bullet - 1)$ on the unit-normalised Avery harmonic $Y_{N_\bullet L_\bullet M_\bullet}^{(3)}$. The joint Lipschitz norm satisfies

$$\|Y_{N_a L_a M_a}^{(3)} \otimes Y_{N_b L_b M_b}^{(3)}\|_{\text{Lip}}^2 = (N_a^2 - 1) \cdot 1 + 1 \cdot (N_b^2 - 1) = N_a^2 + N_b^2 - 2, \quad (21)$$

where we have used $\|Y_{N_a L_a M_a}^{(3)}\|_{\text{Lip}}^2 = N_a^2 - 1$ (the round- S^3 Laplace–Beltrami eigenvalue at level N_a) and $\|Y_{N_a L_a M_a}^{(3)}\|_\infty \leq 1$ on the unit-normalised panel. Combining (19) and (21) gives (20).

The supremum over the irrep rectangle $[2, n_{\max,a} + 1] \times [2, n_{\max,b} + 1]$ in (15) is finite and bounded by $\sqrt{2}$: since $((N_a - 1) + (N_b - 1))^2 \leq 2((N_a - 1)^2 + (N_b - 1)^2) \leq 2(N_a^2 + N_b^2 - 2)$, the per-irrep ratio never exceeds 2. The supremum is computed by scanning the (finite) irrep grid; it equals 1 at (2, 2), exceeds 1 for every larger cutoff, and increases to $\sqrt{2} \approx 1.414$ as $n_{\max,a}, n_{\max,b} \rightarrow \infty$ (diagonal values 1.000 at (2, 2), 1.095 at (3, 3), 1.155 at (4, 4); see Remark 4.7). $\square$

Remark 4.7 (Diagonal closed form). On the diagonal $n_{\max,a} = n_{\max,b} = n$, the supremum in (15) is attained at the corner $(n + 1, n + 1)$ and reads

$$C_3^{(2)}(n, n) = \sqrt{\frac{(2n)^2}{2(n+1)^2 - 2}} = \sqrt{\frac{2n}{n+2}}.$$

Numerical values: $C_3^{(2)}(2, 2) = 1$, $C_3^{(2)}(3, 3) = \sqrt{6/5} \approx 1.095$, $C_3^{(2)}(10, 10) \approx 1.291$, $C_3^{(2)}(100, 100) \approx 1.400$, with $C_3^{(2)}(n, n) \nearrow \sqrt{2}$. The constant exceeds 1 for every $n \geq 3$; the convergence in (14) is nonetheless driven entirely by the factor $\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}) \rightarrow 0$, which the bounded constant $C_3^{(2)} \leq \sqrt{2}$ does not disturb.

Remark 4.8 (Asymmetric supremum). For asymmetric cutoffs $n_{\max,a} \neq n_{\max,b}$, the supremum is again bounded by $\sqrt{2}$ and is found by scanning the irrep grid. At $(n_{\max,a}, n_{\max,b}) = (10, 4)$ for example, the maximum over $[2, 11] \times [2, 5]$ is ≈ 1.179 at the *interior* point (8, 5) ($\sqrt{121/87}$) — the supremum need not occur at the grid corner ((11, 5) gives only $\sqrt{196/144} \approx 1.167$). In all cases $C_3^{(2)} \leq \sqrt{2}$, which is what the convergence theorem requires.

4.4 Lemma L4-T: joint Berezin reconstruction

Definition 4.9 (Joint Berezin reconstruction). For $n_{\max,a}, n_{\max,b} \geq 1$ and $f, g \in C^\infty(S^3)$, define

$$B_{n_{\max,a}, n_{\max,b}}(f \otimes g) := B_{n_{\max,a}}(f) \otimes B_{n_{\max,b}}(g), \quad (22)$$

and extend to general $h \in C^\infty(S^3 \times S^3)$ by linear extension on the dense subspace of finite linear combinations of simple tensors.

Lemma 4.10 (L4-T: joint Berezin properties). *The map $B_{n_{\max,a},n_{\max,b}} : C^\infty(S^3 \times S^3) \rightarrow \mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$ has the following four properties on simple tensors (and extends to positive cones / by density):*

- (a) (Positivity.) *If $f, g \geq 0$ on S^3 , then $B_{n_{\max,a},n_{\max,b}}(f \otimes g)$ is positive semi-definite.*
- (b) (Contractivity.) $\|B_{n_{\max,a},n_{\max,b}}(f \otimes g)\|_{\text{op}} \leq \|f\|_\infty \cdot \|g\|_\infty = \|f \otimes g\|_\infty$.
- (c) (Approximate identity.)

$$\|B_{n_{\max,a},n_{\max,b}}(f \otimes g) - (P_{n_{\max,a}} \otimes P_{n_{\max,b}})M_{f \otimes g}(P_{n_{\max,a}} \otimes P_{n_{\max,b}})\|_{\text{op}} \leq \gamma_{n_{\max,a}} \|g\|_\infty \|\nabla f\|_\infty + \|f\|_\infty \gamma_{n_{\max,b}} \|\nabla g\|_\infty$$

- (d) (Compatibility with L3-T.) $\|[D_{a,b}, B_{n_{\max,a},n_{\max,b}}(f \otimes g)]\|_{\text{op}} \leq C_3^{(2)} \|f \otimes g\|_{\text{Lip}}$, with $C_3^{(2)} \leq \sqrt{2}$ given by (15).

Proof. Each property factorizes from the single-factor property [20, Lemma 3.6]. **(a)** The Kronecker product of two positive semi-definite operators is positive semi-definite. **(b)** $\|B_{n_{\max,a}}(f) \otimes B_{n_{\max,b}}(g)\|_{\text{op}} = \|B_{n_{\max,a}}(f)\|_{\text{op}} \cdot \|B_{n_{\max,b}}(g)\|_{\text{op}} \leq \|f\|_\infty \cdot \|g\|_\infty$ by single-factor L4(b) on each side. **(c)** Use the telescoping identity

$$B_a(f) \otimes B_b(g) - P_a M_f P_a \otimes P_b M_g P_b = (B_a(f) - P_a M_f P_a) \otimes B_b(g) + P_a M_f P_a \otimes (B_b(g) - P_b M_g P_b),$$

take operator norms, and apply single-factor L4(c) on each summand. **(d)** Apply Lemma 4.6 to the simple tensor $B_a(f) \otimes B_b(g) \in \mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$, using single-factor L4(d) for the per-factor commutator bound and the triangle bound (19) for the operator-norm reduction. $\square$

4.5 Lemma L5-T: joint state-space-GH assembly

Lemma 4.11 (L5-T: joint state-space-GH bound from the joint tunneling pair). *The joint tunneling pair $(B_{n_{\max,a},n_{\max,b}}, P_{n_{\max,a}} \otimes P_{n_{\max,b}})$ with $B_{n_{\max,a},n_{\max,b}}$ from Definition 4.9 and $P_{n_{\max,a}} \otimes P_{n_{\max,b}}$ the joint spectral compression is a Latrémolière tunneling pair between $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ and $\mathcal{T}_{n_{\max,a}}^{\lambda_a} \otimes \mathcal{T}_{n_{\max,b}}^{\lambda_b}$, and yields the bound*

$$\Lambda(\mathcal{T}_{n_{\max,a}}^{\lambda_a} \otimes \mathcal{T}_{n_{\max,b}}^{\lambda_b}, \mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}) \leq C_3^{(2)} \cdot (1 + 2\sqrt{2}) \cdot \max\left(\frac{\gamma_{n_{\max,a}}}{\lambda_a}, \frac{\gamma_{n_{\max,b}}}{\lambda_b}\right). \quad (23)$$

Proof. We work at unit focal length $\lambda_a = \lambda_b = 1$ and reinstate the rescaling at the end. The four constituents of a Latrémolière tunnel for metric spectral triples [12] on the joint pair are:

- (i) $\text{reach}_{B_{n_{\max,a},n_{\max,b}}}$;
- (ii) $\text{reach}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}}$;
- (iii) $\text{height}_{B_{n_{\max,a},n_{\max,b}}}$, the Lipschitz-distortion height (Lipschitz-seminorm-comparison form [20, Remark 3.8]);
- (iv) $\text{height}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}} = 0$ for the orthogonal projection.

Reach. By Lemma 4.10(c) on the unit-Lipschitz panel,

$$\text{reach}_{B_{n_{\max,a},n_{\max,b}}} \leq \gamma_{n_{\max,a}} + \gamma_{n_{\max,b}} \leq 2 \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}).$$

The dual $\text{reach}_{P_{n_{\max,a}} \otimes P_{n_{\max,b}}}$ inherits the same bound by symmetry of the joint convolution — and with it the single-factor reach_P *named gap* (the invalid cb-norm dual-reach that descoped Papers 38/40), so the bound below is at the state-space-GH level, not Latrémolière propinquity.

Height. By the Lipschitz-distortion form [12, §4] [20, Remark 3.8],

$$\text{height}_{B_{n_{\max,a}, n_{\max,b}}} = \sup_{\|h\|_{\text{Lip}} \leq 1} \left| \|h\|_{\text{Lip}} - \|B_{n_{\max,a}, n_{\max,b}}(h)\|_{\text{Lip}}^{\mathcal{O}} \right|.$$

Taking the supremum on simple tensors $h = f \otimes g$ with $L_f := \|\nabla f\|_{\infty}$, $M_f := \|f\|_{\infty}$ (and similarly for g) and using $\|f \otimes g\|_{\text{Lip}}^2 \leq L_f^2 M_g^2 + M_f^2 L_g^2$ (the pointwise continuum-gradient identity on the product Riemannian metric — this Pythagorean form is for the *classical* gradient and is exact) together with the triangle bound (19) applied to the joint commutator, single-factor L4(d) on each factor, and the elementary identity $|x - y| = (x^2 - y^2)/(x + y)$ on $\|h\|_{\text{Lip}}^2 - \|B(h)\|_{\text{Lip}}^2$ gives the explicit bound

$$\text{height}_{B_{n_{\max,a}, n_{\max,b}}}(f \otimes g) \leq \frac{2L_f^2 M_g^2 \gamma_{n_{\max,a}} + 2L_g^2 M_f^2 \gamma_{n_{\max,b}}}{\sqrt{L_f^2 M_g^2 + M_f^2 L_g^2}}. \quad (24)$$

On the unit-norm panel ($L_f = L_g = M_f = M_g = 1$), this collapses to

$$\text{height}_{B_{n_{\max,a}, n_{\max,b}}} \leq \sqrt{2}(\gamma_{n_{\max,a}} + \gamma_{n_{\max,b}}) \leq 2\sqrt{2} \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}). \quad (25)$$

The dual height $\text{height}_{P_{n_{\max,a} \otimes P_{n_{\max,b}}}} = 0$ since the joint projection is exact on its image.

Assembly. Combining the reach and height bounds with the $C_3^{(2)}$ Lipschitz comparison constant of Lemma 4.6 (which controls the conversion between $\|\cdot\|_{\text{Lip}}^{\mathcal{O}}$ in the truncated triple and the classical $\|\nabla \cdot\|_{\infty}$ on $S^3 \times S^3$),

$$\begin{aligned} \Lambda &\leq \max(\text{reach}_B, \text{reach}_P, \text{height}_B, \text{height}_P) \leq C_3^{(2)} \cdot (\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}) + 2\sqrt{2} \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})) \\ &= C_3^{(2)} \cdot (1 + 2\sqrt{2}) \cdot \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}). \end{aligned}$$

The λ^{-1} -rescalings of (9) propagate linearly through the bound and produce the $\max(\gamma/\lambda)$ -form of (23). $\square$

Remark 4.12 (Identification of the cross-Stein–Weiss term). The first-pass internal proof of this lemma sketched the cross-term contribution as $\varepsilon_{\text{cross}} = O(\gamma_{n_{\max,a}} \gamma_{n_{\max,b}})$, suggesting a quadratic decay rate. The closed-form analysis above shows the actual rate is linear: $\varepsilon_{\text{cross}} \leq (2\sqrt{2} - 1) \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}}) \approx 1.83 \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$ on the unit-norm panel. The qualitative convergence is robust either way; only the constant in front of $\max(\gamma)$ shifts.

5 Numerical verification

We verify the joint propinquity bound at the panel $(n_{\max,a}, n_{\max,b}) \in \{(2, 2), (2, 3), (3, 3), (3, 4), (4, 4)\}$.

Single-factor input. The single-factor moments $\gamma_{n_{\max}}$ inherit from [20, Lemma 3.4]:

$$\gamma_2 = \pi - \frac{64\sqrt{2}}{27\pi} \approx 2.0746, \quad \gamma_3 \approx 1.6101, \quad \gamma_4 \approx 1.3223.$$

Joint cb-norm. By Lemma 4.3(c), the joint cb-norms are exact rationals: $\|T^{(2,2)}\|_{\text{cb}} = 4/9$, $\|T^{(2,3)}\|_{\text{cb}} = 1/3$, $\|T^{(3,3)}\|_{\text{cb}} = 1/4$, $\|T^{(3,4)}\|_{\text{cb}} = 1/5$, $\|T^{(4,4)}\|_{\text{cb}} = 4/25$.

Joint Lipschitz comparison constant. By Lemma 4.6 (tight triangle bound), the diagonal sequence is

$$C_3^{(2)}(n, n) = \sqrt{2n/(n+2)} \nearrow \sqrt{2},$$

giving $C_3^{(2)}(2, 2) = 1$, $C_3^{(2)}(3, 3) = \sqrt{6/5} \approx 1.095$, $C_3^{(2)}(4, 4) = \sqrt{8/6} \approx 1.155$.

Joint propinquity bound (unit focal length). We report two presentations of the bound at the panel: the *factorised-panel* bound $\Lambda^{\text{fact}} = C_3^{(2),\text{fact}} \cdot \max(\gamma_a, \gamma_b)$ with $C_3^{(2),\text{fact}} = 1$ on the unit-Lipschitz simple-tensor panel (this is the variant computed by `compute_tensor_propinquity_bound` as `propinquity_bound`), and the *full-operator-system* bound (23) obtained by inserting the triangle-bound $C_3^{(2)} \leq \sqrt{2}$ of Lemma 4.6 together with the $1 + 2\sqrt{2}$ reach-plus-height factor of Lemma 4.11.

$(n_{\max,a}, n_{\max,b})$	$\gamma_{n_{\max,a}}$	$\gamma_{n_{\max,b}}$	$\ T^{(a,b)}\ _{\text{cb}}$	Λ^{fact}	$C_3^{(2)}$	Λ^{full}
(2, 2)	2.0746	2.0746	4/9	2.0746	1.000	7.942
(2, 3)	2.0746	1.6101	1/3	2.0746	1.043	8.281
(3, 3)	1.6101	1.6101	1/4	1.6101	1.095	6.752
(3, 4)	1.6101	1.3223	1/5	1.6101	1.121	6.909
(4, 4)	1.3223	1.3223	4/25	1.3223	1.155	5.845

The factorised-panel column Λ^{fact} uses $\Lambda^{\text{fact}} \leq \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$, the value tracked in the supporting computational module as the factorised-panel propinquity bound. The full-operator-system column Λ^{full} uses $\Lambda \leq C_3^{(2)} \cdot (1 + 2\sqrt{2}) \cdot \max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$ with $1 + 2\sqrt{2} \approx 3.828$ (computed from this closed form; the supporting module’s `propinquity_bound_full` field tracks a looser conservative 2γ estimate, not this assembled bound). Both columns vanish as $n_{\max,a}, n_{\max,b} \rightarrow \infty$; both exhibit monotone decrease across the diagonal panel.

Three structural observations.

1. *Monotone decrease and bit-identical match to single factor.* On the factorised panel, $\Lambda^{\text{fact}}(2, 2) = 2.0746$, $\Lambda^{\text{fact}}(3, 3) = 1.6101$, $\Lambda^{\text{fact}}(4, 4) = 1.3223$. The diagonal ratio $\Lambda^{\text{fact}}(4, 4)/\Lambda^{\text{fact}}(2, 2) = 1.3223/2.0746 = 0.6374$ matches the single-factor Paper 38 ratio $\Lambda^{(1)}(4)/\Lambda^{(1)}(2) \approx 0.637$ *bit-identically* to four digits; the joint max-rate is dominated by the slower factor and inherits the single-factor cutoff scaling. On the full-operator-system column the analogous ratio is $\Lambda^{\text{full}}(4, 4)/\Lambda^{\text{full}}(2, 2) \approx 5.845/7.942 \approx 0.736$, with the (now increasing) $C_3^{(2)}$ factor partially offsetting the $\max(\gamma)$ decrease.
2. *Joint cb-norm decreases as $1/(n_{\max,a}n_{\max,b})$.* $\|T^{(2,2)}\|_{\text{cb}} = 4/9 \approx 0.444$ vs. $\|T^{(4,4)}\|_{\text{cb}} = 4/25 = 0.16$, an exact factor of $9/25 = 0.36$ reduction matching the closed form of Lemma 4.3(c). The asymptotic order $O(1/(n_{\max,a}n_{\max,b}))$ is one order faster than the single-factor cb-norm $O(1/n_{\max})$.
3. *Test count.* The supporting computational module `geovac/gh_convergence_tensor.py` (~ 1881 lines) and the test file `tests/test_gh_convergence_tensor.py` contain the full verification panel, including `tests/test_paper39_triangle_tight.py`, which builds the joint commutator from the real Camporesi–Higuchi matrices and the graded multipliers and confirms (i) the triangle bound (19) is tight (ratio $\rightarrow 1$) and (ii) the Pythagorean form is maximally violated (ratio $\rightarrow 2$), the computational basis for the $C_3^{(2)} \leq \sqrt{2}$ constant.

6 Related work and significance

6.1 Place in the Connes–van Suijlekom / Latrémolière lineage

The published propinquity tensor-product literature, as discussed in §1, is summarized in the following table.

Setup	Status	Reference
Two finite-dimensional	Standard	Connes–Chamseddine AC
Riemannian $\times$ finite	Closed	Latrémoière [14]
$C(X) \times A$ (A AF)	Closed	Aguilar [1]
Riemannian $\times$ abelian	Closed	Farsi–Latrémoière [7]
Single Riemannian along path	Closed	Farsi–Latrémoière [8]
Single truncation, NC integral	Closed	Hekkelman–McDonald [10]
Two infinite Riemannian non-flat	Open prior to this work	—
Two S^3 at distinct λ	Closed (this paper)	Theorem 3.1

The structural difficulty of the two-infinite case is that Latrémoière tunneling pairs do not multiply trivially: for $(B_a \otimes B_b, P_a \otimes P_b)$, the joint reach inherits from both single-factor approximate-identity rates with their commutator structure, and the joint Lipschitz constant is not $\max(C_3^a, C_3^b)$ in general. The Bożejko–Fendler product cb-norm formula covers each factor’s central convolution but the propinquity-tunneling assembly for the product is not in print prior to this work. The present paper closes the case by explicitly using the Connes–Marcolli graded anticommutation (12) on the KO-dim-6 module to produce the clean two-term graded Leibniz rule, whose joint commutator is then controlled — with constant $C_3^{(2)} \leq \sqrt{2}$ — by the (tight) operator-norm triangle bound; this controls both the joint Lipschitz constant (Lemma 4.6) and the joint height-distortion term (Lemma 4.11).

6.2 Obstructions to the cross-manifold extensions

We documented in §1 that the same-manifold case treated here is the most accessible of three plausible tensor-product extensions of [20]. The two cross-manifold variants remain open structural questions in the broader noncommutative geometry program.

Riemannian $\times$ Hardy-sector ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Bargmann}}$). The first cross-manifold extension would tensor the Riemannian S^3 Camporesi–Higuchi triple with a Bargmann–Segal-type spectral triple on S^5 in the holomorphic Hardy sector [19]. This is structurally obstructed at the framework level: the S^3 factor is a second-order Riemannian spinor Dirac with KO-dimension 3, while the S^5 Bargmann factor is a first-order complex-analytic Euler operator on a Hardy sector and does not fit standard NCG Dirac axioms (KO-dimension is not classically defined in this setting). The two factors live in different categories of NCG construction, and there is no published prescription for tensoring them. We have flagged this elsewhere as the four-layer Coulomb/HO asymmetry of [19] §V resurfacing at the spectral-triple level. Neither the Bargmann transform nor the Kustaanheimo–Stiefel transformation provides a bridge: the former is a unitary intertwiner of operators on Hilbert spaces, not a unitary equivalence of spectral triples, and the latter operates at the coordinate level, mapping the 3D Coulomb problem to a 4D HO problem under a constraint, without preserving the Riemannian / spinor bundle structure required for KO-dimension. A proper Riemannian–Hardy propinquity theory is not in the literature and would require new framework development beyond the scope of the present paper.

Riemannian $\times$ Riemannian on different manifolds ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Riem}}$). The alternative is to abandon the Bargmann content of S^5 and tensor S^3 with the round Riemannian S^5 spinor Dirac. KO-arithmetic is clean ($3 + 5 = 8 \equiv 0 \pmod{8}$), and this object is a textbook Connes–Marcolli construction. But propinquity convergence for $\mathcal{T}_{S^5}^{\text{Riem}}$ alone — the rank-2 Spin(6) analog of [20] — requires a fresh five-lemma analysis: a higher-rank version of the Avery 3-Y integral, the corresponding SO(6) selection rule, and a rank-2 central spectral Fejér kernel. We expect analogous $\log n/n$ -rate convergence with rank-dependent constants, but the detailed analysis is open and is itself a separate paper.

6.3 Significance for the multi-focal-composition program

The same-manifold case is what is needed for the GeoVac multi-focal composition program (where two electrons in distinct hydrogenic shells live on S^3 at distinct focal lengths $\lambda = n/Z$). With Theorem 3.1 in place, the cross-register two-body coordinate operator $V(\hat{r}_1, \hat{r}_2)$ across two electronic registers at distinct focal lengths is well-defined as a multiplier on the joint operator system $\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$, with Lipschitz seminorm bounded by the product of $C_3^{(2)}(n_{\max,a}, n_{\max,b})$ and the classical Lipschitz norm of V on $S^3 \times S^3$. The convergence theorem provides the metric foundation; the structural application is left for a separate paper.

7 Open questions

(i) **Quantitative rate at small $n_{\max}$.** Theorem 3.1 inherits the single-factor asymptotic rate $\gamma_n \sim (4/\pi) \log n/n$ from [20]; the quantitative rate at small $n_{\max}$ remains the open follow-up to [20, §5(i)], parallel to single-factor Track C in the GeoVac internal record. The joint rate inherits the same open-quantitative-constant feature, since the max-form $\max(\gamma_{n_{\max,a}}, \gamma_{n_{\max,b}})$ is dominated by the worse single-factor rate.

(ii) **Off-factorized-panel functions.** Theorem 3.1 is stated and verified on simple tensors $f \otimes g$ and extended by linearity to finite linear combinations; the $C_3^{(2)}$ formula (15) arises from the (tight) triangle bound on the graded Leibniz rule for unit-Lipschitz Avery harmonics, and is sharp on factorized harmonics ($C_3^{(2)} = \sqrt{2}$ is achieved). General $h \in C^\infty(S^3 \times S^3)$ that are not of factorized form may admit sharper bounds via direct Sobolev embedding on $S^3 \times S^3$. The detailed analysis of the off-factorized case is open.

(iii) **Higher tensor factors — structural extension (sketch; the $k = 2$ case is the theorem of this paper).** The proof extends mechanically to k -fold tensor products $\mathcal{T}_{S^3}^{\lambda_1} \otimes \cdots \otimes \mathcal{T}_{S^3}^{\lambda_k}$ via repeated application of the graded Leibniz rule and the operator-norm triangle bound:

$$\Lambda^{(k)}(\mathcal{T}_{n_1, \dots, n_k}^{(k)}, \mathcal{T}_{(S^3)^{\otimes k}}^{(k)}) \leq C_3^{(k)}(n_1, \dots, n_k) \cdot \gamma_{n_1, \dots, n_k}^{(k)}, \quad (26)$$

with the triangle bound

$$C_3^{(k)}(n_1, \dots, n_k) \leq \sqrt{k} \sqrt{\frac{N_* - 1}{N_* + 1}} \nearrow \sqrt{k}, \quad N_* := \max_j n_j, \quad (27)$$

and joint rate $\gamma^{(k)} = \max_j \gamma_{n_j}^{\text{SU}(2)}$ (worst-factor SU(2) rate). *Correction (2026-06-18):* an earlier draft claimed a **k -independent** constant $\sqrt{(N_* - 1)/(N_* + 1)} \rightarrow 1^-$ via a “Pythagorean Leibniz” sum-of-squares bound $\|\sum_j X_j\|_{\text{op}}^2 = \sum_j \|X_j\|_{\text{op}}^2$. That identity is **false**: pairwise anti-commutation of the $D^{(k)}$ summands does not make the resulting commutator terms operator-norm orthogonal (Remark 4.5; verified maximally violated at $k = 2$). The correct bound is the triangle one, which gives a genuine $\sqrt{k}$ factor (not eliminated, as had been claimed) from $(\sum_j (N_j - 1))^2 \leq k \sum_j (N_j - 1)^2$, hence $C_3^{(k)} \nearrow \sqrt{k}$. For each *fixed* k this is a finite constant, so convergence (26) still holds (driven by $\gamma^{(k)} \rightarrow 0$); but the constant grows with the number of factors and is *not* a k -independent drop-in tool. This $k > 2$ extension is a sketch: only the $k = 2$ case is proved and verified in this paper.

(iv) **Conditional Coulomb/HO bridge.** Should a unifying NCG-framework extension be developed that bridges the Riemannian spinor Dirac and Hardy-sector first-order complex operator categories at the spectral-triple level (§6.2), the W2b-medium case $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Bargmann}}$ would become a target for a follow-up to the present paper. The ingredients on the S^3 side are exactly Theorem 3.1 restricted to the left tensor factor.

(v) **Tensor products at higher rank — structural extension (sketch).** The present paper treats the equal-manifold tensor product $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ at two distinct focal lengths on the same compact group $SU(2)$. A companion higher-rank generalisation of the single-factor result of [20] has since been established in [21], which lifts Theorem 3.1 of [20] to all compact connected Lie groups G with bi-invariant metric and proves that the rate constant $4/\pi$ is *universal* across the class via a Plancherel-weight $\times$ Vandermonde-Jacobian cancellation theorem.

Combining the k -fold extension (item (iii) above) with the general- G ingredients of [21] gives the expected multi-factor form

$$\Lambda^{(k)}(\mathcal{T}_{n_1, \dots, n_k}^{(k)}, \mathcal{T}_{G_1 \times \dots \times G_k}^{(k)}) \leq C_3^{(k)} \cdot \max_j \gamma_{n_j}^{G_j}, \quad C_3^{(k)} \leq \sqrt{k} \sup_j C_3^{G_j}(n_j) \nearrow \sqrt{k}, \quad (28)$$

for any $k \geq 1$ and arbitrary compact connected Lie groups $G_1, \dots, G_k$ with bi-invariant metrics, where $C_3^{G_j}(n_j) \rightarrow 1^-$ is the Paper 40 [21] per-factor L3 constant and $\gamma_{n_j}^{G_j}$ is the per-factor general- G rate at the universal $(4/\pi) \log n_j/n_j$ asymptote. *Correction (2026-06-18):* an earlier draft boxed this as a closed “master theorem” with a **k -independent** constant $\sup_j C_3^{G_j}(n_j)$; that rested on the withdrawn Pythagorean sum-of-squares bound (item (iii)). With the correct triangle bound the constant carries a genuine $\sqrt{k}$ factor and is *not* k -independent. For each fixed k convergence holds (finite constant), and the joint rate still inherits the universal $4/\pi$ asymptotic from [21]; but this multi-factor / general- G form is a **sketch** (only the $k = 2$, $SU(2)$ case is proved here, and the $k = 1$ cases in [20, 21]), not a separately verified master theorem.

Practical applications: Paper 45 [22]’s compact-temporal carrier $S^3 \times S^1$ instantiates the $k = 2$, $G_1 = SU(2)$, $G_2 = U(1)$ case of (28) at the level of its signature-agnostic product-carrier rate. Note, however, that the Lorentzian (Krein) reading of that carrier does *not* yield propinquity convergence: Paper 45 proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically), so the master construction provides the surviving Riemannian product-carrier rate, not a Lorentzian convergence result.

8 Conclusion

We have established van Suijlekom state-space Gromov–Hausdorff convergence for truncations of the tensor-product Camporesi–Higuchi spectral triple on $S^3 \times S^3$ at distinct focal lengths (the strictly stronger Latrémolière propinquity is not claimed; see the convergence convention of §2), with explicit qualitative-rate bound (3), closed-form joint Lipschitz comparison constant (15) bounded by $\sqrt{2}$ at every finite cutoff, and explicit Stein–Weiss-type bound on the Lipschitz-distortion height. The mechanism that controls both the Lipschitz constant and the height term is the same: the Connes–Marcolli graded anticommutation $\{D_{CH}^{(a)} \otimes I, \gamma_a \otimes D_{CH}^{(b)}\} = 0$ on the KO-dimension-6 graded module produces a clean two-term graded Leibniz rule, whose joint commutator is controlled by the (tight) operator-norm triangle bound $\|A + B\| \leq \|A\| + \|B\|$ (an earlier Pythagorean refinement is withdrawn — see Remark 4.5 — and convergence is unaffected since $C_3^{(2)} \leq \sqrt{2}$ is a finite constant). The result extends Connes–van Suijlekom [6] and the subsequent flat-structure and single-non-abelian-factor literature [9, 10, 15, 20] to the simplest non-abelian tensor case, and closes a case in the propinquity tensor-product programme that the published literature does not previously cover. The cross-manifold extensions remain open structural questions, with the Coulomb/HO category mismatch documented in [19] §V identifying a precise framework-level obstruction to the Riemannian–Hardy variant.

Acknowledgments

The author thanks Alain Connes, Walter van Suijlekom, François Latrémolière, Carla Farsi, Matilde Marcolli, Eva-Maria Hekkelman, Edward McDonald, Malte Leimbach, and Carlos Perez-

Sanchez for the foundational frameworks on which this paper builds; remaining errors are the author's own. This paper builds directly on [20] (single-factor S^3 state-space GH convergence) and on the multi-focal-composition track of the GeoVac internal record. As with [20], this work was developed in an AI-augmented agentic workflow with Anthropic's Claude: implementation of the proof-assistant verification machinery, drafting of the internal proof memos used as supporting documentation, and bibliographic collation were performed collaboratively. All theorem statements, design choices, and editorial decisions are the author's responsibility.

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- [21] J. Loutey, *State-space Gromov–Hausdorff convergence of spectral truncations of compact Lie groups with bi-invariant metric*, preprint, May 2026; higher-rank single-factor companion to the present tensor-product paper, DOI via Zenodo. Generalizes [20] from $SU(2)$ to all compact connected Lie groups; rate constant $4/\pi$ universal across the class via a Plancherel-weight $\times$ Vandermonde-Jacobian cancellation theorem; asymptotic-tight $C_3 = 1$ at all ranks via the PRV-summand bound of Kumar (1988) and Vinberg (1990).
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